

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (EE/ME/CL/CE/IT/EC)

May 2013

PY 101 Engineering Physics

Date: 08.05.2013, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Choose the correct answer for the following questions.

[07]

- (a) The frequency of a wave travelling at a speed of 500 ms^{-1} is 25 Hz . Write the wavelength with due significant figures.
- (i) 20 mm (ii) 2.0 mm
(iii) 20.0 mm (iv) 0.20 m
- (b) If $C = 2.5 \pm 0.5 \text{ m/s}$, $D = 0.10 \pm 0.01 \text{ s}$ what is $C \times D$?
- (i) $0.25 \pm 0.05 \text{ m}$ (ii) $0.25 \pm 0.08 \text{ m}$
(iii) $0.25 \pm 0.51 \text{ m}$ (iv) $0.25 \pm 0.25 \text{ m}$
- (c) What is the value of intensity level of threshold of feeling?
- (i) 120 dB (ii) 60 dB
(iii) 20 dB (iv) 100 dB
- (d) The persistence of audible sound due to the successive reflections from the surrounding objects even after the source has stopped to produce that sound is called _____.
- (i) reflection (ii) reverberation
(iii) echo (iv) none of the above
- (e) A property of sound which is most closely associated with the pitch of a musical note is _____.
- (i) amplitude (ii) frequency
(iii) velocity (iv) none of the above
- (f) A train whistle has an acoustic power output of 100 W . If the sound energy spreads out spherically, what is the intensity level in dB at a distance of 100 m from the train? (Given that $I_0 = 10^{-12} \text{ W m}^{-2}$)
- (i) 55 dB (ii) 95 dB
(iii) 89 dB (iv) 120 dB
- (g) Which of the following laser emits light of wavelength in infrared range?

(i) Nd:YAG laser

(ii) CO₂ laser

(iii) Ruby laser

(iv) Both (a) and (b)

- Q - 2 (a) What are the possible sources of systematic errors in the experiment to measure the velocity of ultrasound in liquid using ultrasonic interferometer? [3]
- (b) Define absorption coefficient. Calculate the total absorption coefficient of cinema hall, whose volume is 8000m^3 and reverberation time 1.8 sec. [4]
- (c) Distinguish between ultrasonic sounds and infrasonic sound. Explain inverse-piezoelectric effect to produce ultrasound with proper diagram. What are the advantages and disadvantages of this method? [7]

OR

- Q - 2 (a) Define standard deviation, standard error and probable error. [3]
- (b) Explain four scientific and engineering applications of ultrasonic waves. [4]
- (c) Discuss the basic requirement for an acoustically good auditorium. [7]
- Q - 3 (a) Calculate the refractive indices of the core and cladding material of a fiber from the following data: $NA = 0.22$ and $\Delta = 0.012$. [3]
- (b) Write any four applications of optical fiber. [4]
- (c) What do you understand by solid state laser? Describe the principle, construction, and working of a solid laser. [7]

OR

- Q - 3 (a) Discuss s significant characteristics of laser beam. [3]
- (b) A three level laser emits a light of wavelength of $5500 \text{ \AA}$, what will be the ratio of population of upper level (E_2) to the lower energy level (E_1) if the optical pumping mechanism is shut off (use $T = 300\text{K}$, $h = 6.63 \times 10^{-34} \text{ J/sec}$, $k_B = 8.62 \times 10^{-5} \text{ eV/K}$). [4]
- (c) What is an optical fiber? Define and explain the terms: (i) acceptance angle, (ii) numerical aperture, (iii) relative refractive index differences, (iv) index profile and (v) propagating modes [7]

SECTION - II

- Q - 4 Choose the correct answer for the following questions. [07]
- (a) The energy gap between the valance and conduction band in a semiconductor is of the order of :
(i) 100 eV (ii) 0.001 eV
(iii) 1.0 eV (iv) 10 eV
- (b) Resistance of a substance in superconducting state becomes:
(i) infinite (ii) finite
(iii) zero (iv) none of these
- (c) Which of the following types of electromagnetic radiation has the highest wavelength?
(i) X-rays (ii) visible light
(iii) ultraviolet (iv) gamma-rays

- (d) What is the operating voltage of the X-ray tube to produce X-rays of wavelength $0.2475 \text{ \AA}$?
- (i) 25 kV (ii) 500 V
 (iii) 50 kV (iv) 100 kV
- (e) The number of Bravais lattice in triclinic system is ____.
- (i) 1 (ii) 2
 (iii) 3 (iv) 4
- (f) If X-rays of wavelength $0.5 \text{ \AA}$ are detected at an angle of 5° in the first order, what is the spacing between the adjacent planes of the crystal?
- (i) $2.86 \text{ \AA}$ (ii) $2.32 \text{ \AA}$
 (iii) $2.46 \text{ \AA}$ (iv) $2.68 \text{ \AA}$
- (g) The Miller indices of a set of parallel planes which make equal intercepts on the three axes are:
- (i) (001) (ii) (010)
 (iii) (100) (iv) (222)

- Q - 5 (a) An X-ray tube is operated at 13.6 kV . Find the maximum speed of electron striking the target. [3]
- (b) Explain the differences in origin of X-rays and visible light. [4]
- (c) What is superconductivity? Explain the distinction between soft and hard superconductors. [7]

OR

- Q - 5 (a) Define and explain the terms: (i) SQUID and (ii) Hall coefficient [3]
- (b) Define and explain Meissner effect in superconductors. [4]
- (c) What is Hall effect? Give an experimental method of calculating the carrier concentration and mobility in a given semiconductor. [7]
- Q - 6 (a) Draw the Miller indices planes (100), (101) and (111) in a cubic unit cell. [3]
- (b) Derive the expression for the interplanar spacing between two parallel planes with Miller indices (hkl) for a cubic lattice of lattice constant a . [4]
- (c) What is nanoscience? Explain any four characteristic properties used to explore objects in the nanoscale? [7]

OR

- Q - 6 (a) Calculate the glancing angle on the cubic face (100) of a rock salt crystal ($a = 2.184 \text{ \AA}$) corresponding to second order reflection of X-rays of wavelength $0.710 \text{ \AA}$. [3]
- (b) Write the unit cell parameters of cubic, tetragonal, orthorhombic and monoclinic crystal and draw the crystal systems. [4]
- (c) Write a short note on nanotechnology. Explain any four applications of nanomaterials in the nanotechnology. [7]
